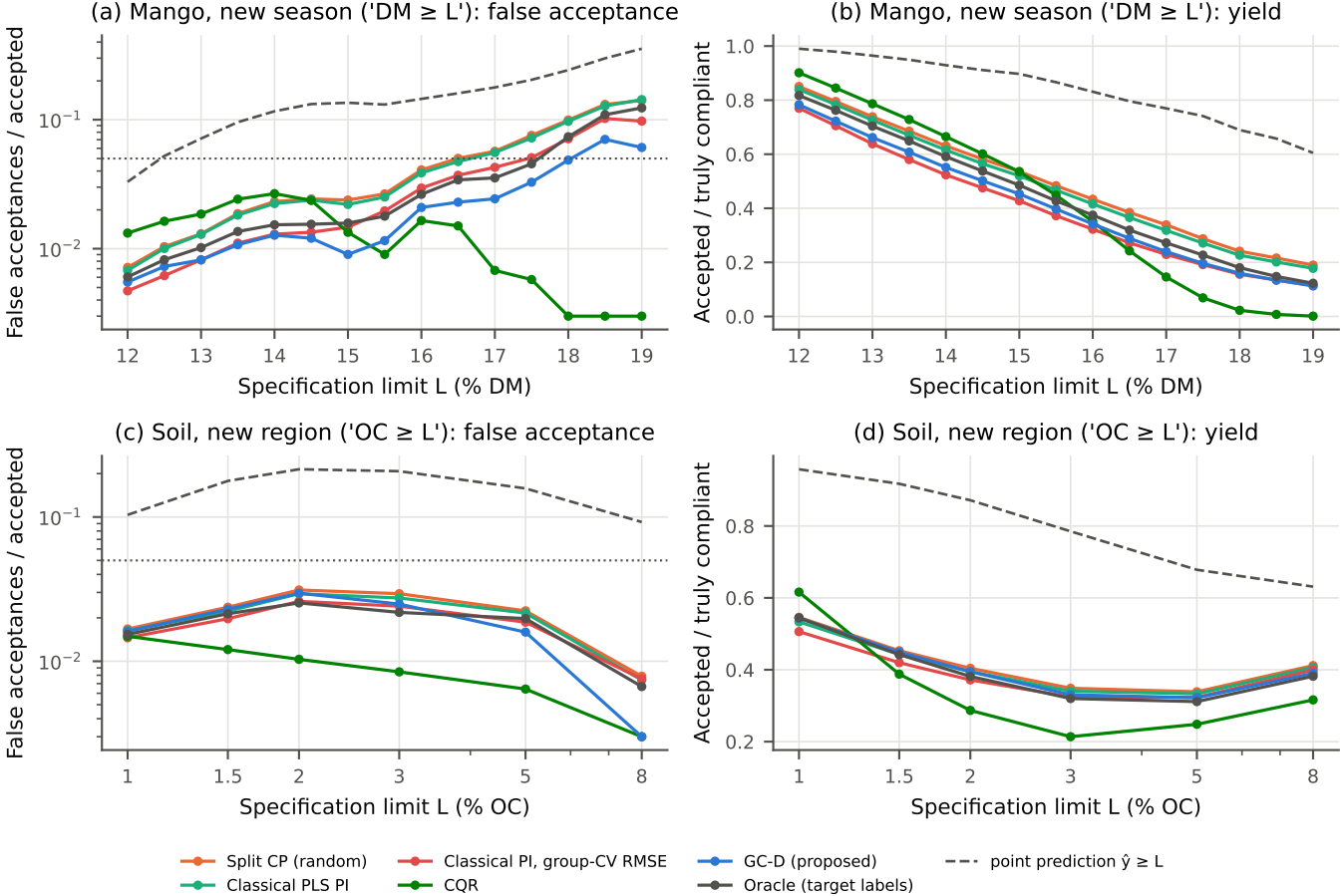
(a) Mango, new season ('DM ≥ L'): false acceptance
False acceptances / accepted
10^{-1}
10^{-2}
12
13
14
15
16
17
18
19
Specification limit L (% DM)
(b) Mango, new season ('DM ≥ L'): yield
Accepted / truly compliant
1.0
0.8
0.6
0.4
0.2
0.0
12
13
14
15
16
17
18
19
Specification limit L (% DM)
(c) Soil, new region ('OC ≥ L'): false acceptance
False acceptances / accepted
10^{-1}
10^{-2}
1
1.5
2
3
5
8
Specification limit L (% OC)
(d) Soil, new region ('OC ≥ L'): yield
Accepted / truly compliant
0.8
0.6
0.4
0.2
1
1.5
2
3
5
8
Specification limit L (% OC)
Split CP (random)
Classical PLS PI
Classical PI, group-CV RMSE
CQR
GC-D (proposed)
Oracle (target labels)
point prediction ŷ ≥ L